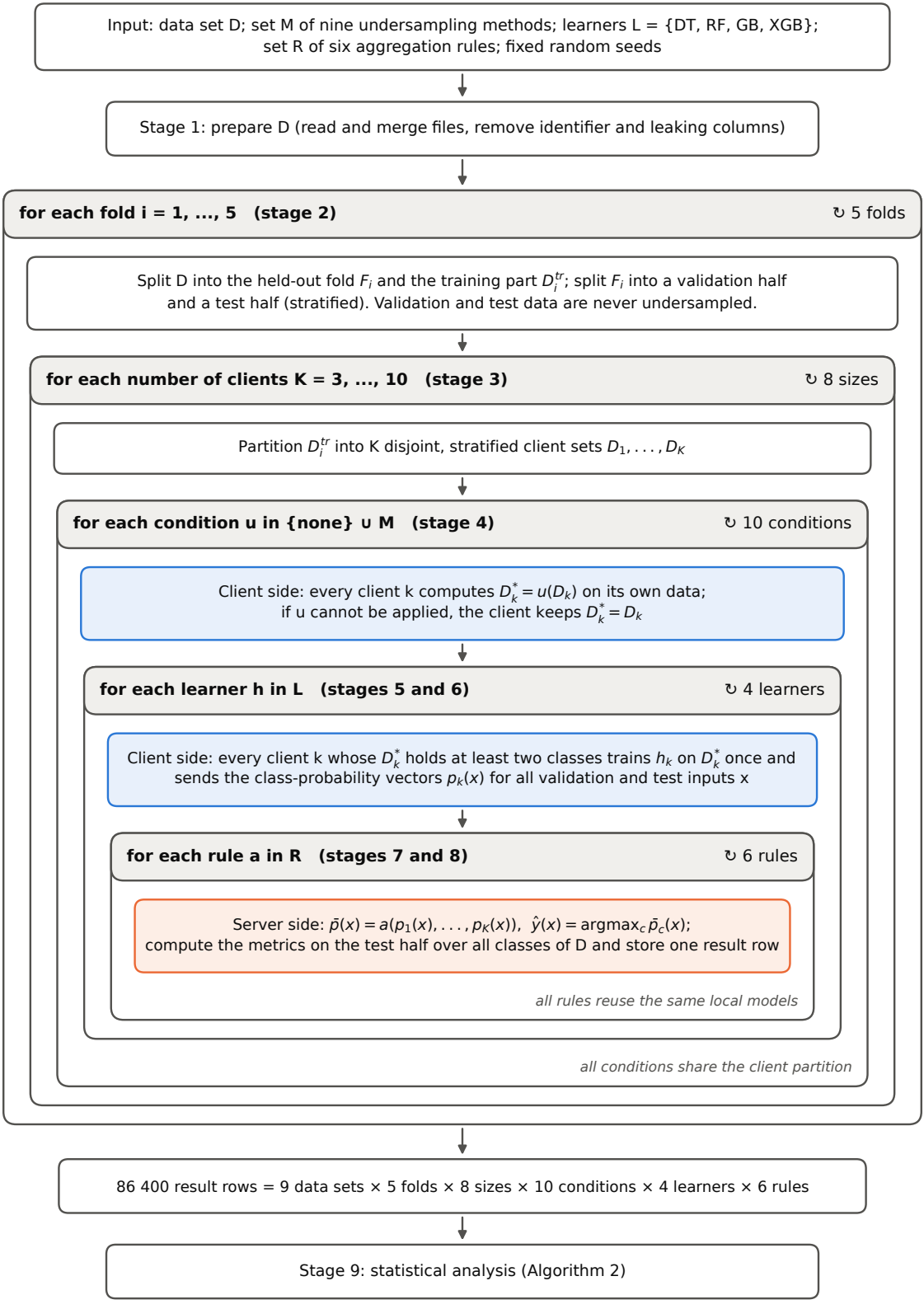
Input: data set D; set M of nine undersampling methods; learners L = {DT, RF, GB, XGB};
set R of six aggregation rules; fixed random seeds
Stage 1: prepare D (read and merge files, remove identifier and leaking columns)
for each fold i = 1, ..., 5 (stage 2)
↻ 5 folds
Split D into the held-out fold F_i and the training part D_i^{tr}; split F_i into a validation half
and a test half (stratified). Validation and test data are never undersampled.
for each number of clients K = 3, ..., 10 (stage 3)
↻ 8 sizes
Partition D_i^{tr} into K disjoint, stratified client sets $D_1, \ldots, D_K$
for each condition u in {none} ∪ M (stage 4)
↻ 10 conditions
Client side: every client k computes $D_k^* = u(D_k)$ on its own data;
if u cannot be applied, the client keeps $D_k^* = D_k$
for each learner h in L (stages 5 and 6)
↻ 4 learners
Client side: every client k whose D_k^* holds at least two classes trains h_k on D_k^* once and
sends the class-probability vectors $p_k(x)$ for all validation and test inputs x
for each rule a in R (stages 7 and 8)
↻ 6 rules
Server side: $\bar{p}(x) = a(p_1(x), \ldots, p_K(x))$, $\hat{y}(x) = \mathrm{argmax}_c \bar{p}_c(x)$;
compute the metrics on the test half over all classes of D and store one result row
all rules reuse the same local models
all conditions share the client partition
86 400 result rows = 9 data sets × 5 folds × 8 sizes × 10 conditions × 4 learners × 6 rules
Stage 9: statistical analysis (Algorithm 2)